

Environmental Product Declaration

In accordance with ISO 14025 and
EN15804:2012+A2:2019 for

Average Steel Slab from Tata Steel Limited



Programme:	The International EPD® System, www.environdec.com
Programme operator:	EPD International AB
Licensee	EPD India
EPD registration number:	EPD-IES-0021266
Version date:	2025-04-20
Validity date:	2030-04-19

An EPD may be updated or depublished if conditions change. To find the latest version of the EPD and to confirm its validity, see www.environdec.com.



General Information

Programme Information

Programme:	The International EPD® System	Licensee: EPD India
Address:	EPD International AB Box 210 60, SE-100 31 Stockholm Sweden	EPD India, a licensee of the International EPD® System 422, Midas, Sahar Plaza Mumbai, India-400059
Website:	www.environdec.com	www.environdecindia.com
E-mail:	info@environdec.com	info@environdecindia.com

Accountabilities for PCR, LCA and independent, third-party verification

Product Category Rules (PCR)

CEN standard EN 15804 serves as the Core Product Category Rules (PCR)

Product Category Rules (PCR):	<i>PCR 2019:14 Construction products, version 1.3.4 Published on 2024.04.30. Based on CEN standard EN 15804. CEN standard EN 15804 serve as the core PCR. UN CPC code 412</i>
PCR review was conducted by:	<i>The Technical Committee of the International EPD®System. See https://www.environdec.com/about-us/the-international-epd-system-about-the-system for a list of members.</i>

Life Cycle Assessment (LCA)

LCA accountability:	Tata Steel Limited
LCA and EPD developer:	Tata Steel Limited

Third-party verification

Independent third-party verification of the declaration and data, according to ISO 14025:2006, via	☒ EPD verification by individual verifier
Third-party verifier:	<i>Dr. Hüdaï Kara, Metsims Sustainability Consulting, 4 Clear Water Place, Oxford OX2 7NL, UK</i> <i>Email: hudai.kara@metsims.com</i>
Approved by:	The International EPD® System
Procedure for follow-up of data during EPD validity involves third party verifier:	
☐ Yes ☒ No	

The EPD owner has the sole ownership, liability, and responsibility for the EPD. EPDs within the same product category but registered in different EPD programmes, or not compliant with EN 15804, may not be comparable. For two EPDs to be comparable, they must be based on the same PCR (including the same version number) or be based on fully-aligned PCRs or versions of PCRs; cover products with identical functions, technical performances and use (e.g. identical declared/functional units);

have equivalent system boundaries and descriptions of data; apply equivalent data quality requirements, methods of data collection, and allocation methods; apply identical cut-off rules and impact assessment methods (including the same version of characterisation factors); have equivalent content declarations; and be valid at the time of comparison. For further information about comparability, see EN 15804 and ISO 14025.

Company Information

Owner of the EPD:	Tata Steel Limited
Contact	Adity Ganguly
E-mail:	adity.ganguly@tatasteel.com

Description of the organization

Tata Steel Limited a flagship company of Tata Group is a multinational steel-making company headquartered in Mumbai (Maharashtra, India). The company is one of the world's most geographically diversified steel producing companies. It was established in India as Asia's first integrated private steel company in 1907 and today the company, together with its subsidiaries, associates and joint ventures, has its presence across five continents with key operations in India, the Netherlands and the United Kingdom. Tata Steel is one of the most diversified integrated steel producers in the world, with an annual crude steel production capacity of 35 MTPA. The Company's manufacturing assets are spread across India, the Netherlands, the UK, and Thailand. In India,

Tata Steel operates in Jamshedpur and Gamharia in Jharkhand, as well as in Kalinganagar and Meramandali in Odisha, with an overall capacity of 21.6 MTPA. Additionally, the Company has downstream manufacturing facilities for wires, tubes, bearings, agricultural equipment, pig iron, ductile iron (DI) pipes, and various industrial by-products. The Company's operations also include Tata Growth Shop, a heavy-duty engineering and fabrication business, along with a Ferro Alloys and Minerals Division. The company's largest steel plant is located in Jamshedpur, Jharkhand (India). The company's operations in India are fully integrated from mining to finished steel production and it also has captive iron ore and coking coal mines in India.



Figure: Tata Steel Limited, Jamshedpur

Product-related or management system-related certifications

All facilities at Tata Steel Limited are ISO 9001, ISO 14001 & ISO 45001 certified.

Name and location of production site(s)

Name	Location
Tata Steel Limited - Jamshedpur	Bistupur, Jamshedpur, Jharkhand – 831001, India
Tata Steel Limited - Kalinganagar	Kalinga Nagar, Gobara Ghati, Jajpur, Odisha 755026, India
Tata Steel Limited - Meramandali	Talabahal, Meramandali, Odisha 759128, India

Product Information

Product name	Average Steel slab
Product identification	Tata Steel Limited has a capacity of 20.12 MTPA crude steel production in India. It meets a variety of standards & offers customizable sizes of cut to length, cut to size, and cut to sheet to cater to the critical needs of the industry. Steel Slabs are also directly supplied to various customers.
Product description	Steel slab manufactured by Tata Steel Ltd in the above facilities represent 100% of total steel slab of Tata Steel Ltd manufactured in India. The technology available in the complex ensures production of high-quality slabs which is then sent to downstream for production of additional products used in critical applications such as automotive, white goods, construction, etc.
UN CPC code	412
Geographical scope	Global

Manufacturing process

The technology used for steel slab production is Blast Furnace (BF) & DRI with Basic Oxygen Furnace (BOF) route comprising older to modern technology. Steel slabs (semi-finished product) are produced from slab caster which takes liquid steel produced from basic oxygen furnace as its raw material. Liquid steel is produced from hot metal in basic oxygen furnace and hot metal is produced in blast furnace.

In this BF-BOF technology, iron ore in the form of hematite and magnetite is reduced in blast furnace through the addition of carbon source which is of internally produced coke and some purchased quantities of metallurgical coke and pulverized coal. Gangue or impurities in the form of alumina and silica present in the iron ore is removed as blast furnace slag by the addition of limestone, pyroxenite and quartz. Hot metal is the primary output of the blast furnace produced along with the generation of blast furnace gas and blast furnace slag. While BF slag is processed further externally to produce Ground Granulated Blast Furnace Slag (GGBS) that gets used in cement production and construction applications, BF gas generated at steel works is predominantly used within the works for heating and power generation applications. Hot metal produced from blast furnace is collected in torpedo ladles and sent to steel melting shops. The molten iron from the blast furnace is transferred to the BOF.

In Tata steel, Meramendali site has coal based DRI whose output is then fed into downstream processes. Iron ore (in the form of lumps or pellets), non-coking coal, and dolomite are the primary raw materials. The iron ore is sized and graded to ensure optimal reduction, while the coal serves as both a fuel and a reducing agent. The raw materials are fed into a rotary kiln, a cylindrical furnace that rotates slowly. Inside the kiln, the materials are heated to temperatures so that the coal combusts, generating carbon monoxide and hydrogen, which act as reducing agents. These gases react with the iron ore, reducing it to metallic iron without melting it. The reduced iron, now in a solid state, forms a porous structure known as sponge iron or DRI.

Hot Metal is sent to steelmaking shops in BOF where it is combined with scrap steel and high-purity oxygen is blown into the mixture at high speeds. This oxidizes impurities such as carbon, silicon, and phosphorus, generating heat and converting the molten iron into steel. Lime is added into the system to remove the oxides formed due to blowing of oxygen. The molten steel is further refined to achieve the desired chemical composition and temperature. This may involve processes such as alloying, degassing, and desulfurization. The refined molten steel is then poured into a continuous casting machine, where it solidifies into semi-finished products called slabs. These slabs are typically rectangular in shape and serve as the primary input for rolling mills.

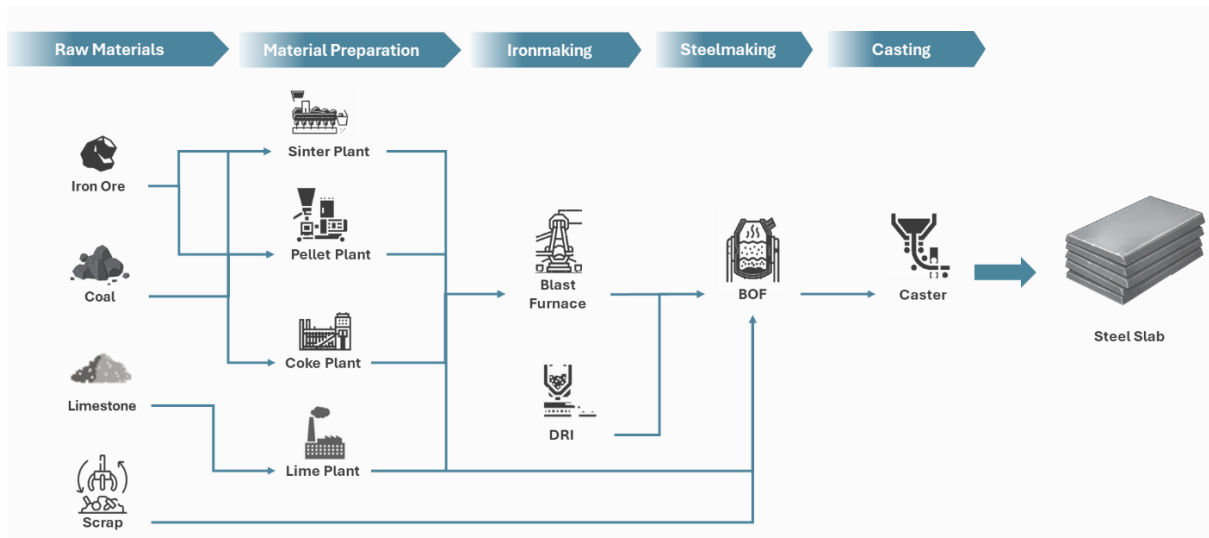


Figure: Process overview from raw materials to Steel Slab

Product Application

A steel slab is a fundamental raw material in the steel manufacturing process, known for its versatility and strength. These slabs are typically rectangular in shape and are produced through the continuous casting of molten steel. They serve as the primary input for rolling mills, where they are further processed into various steel products such as sheets, plates, and coils. Steel slabs are prized for their durability, high tensile strength, and resistance to corrosion,

making them ideal for use in construction, automotive, shipbuilding, and numerous other industries. Steel slabs are semi-finished product & a key raw material for hot rolled coils & mostly used in a variety of applications like Automobiles, Boiler and Pressure Vessels, Ship Building, Railways, Transmission Towers, Oil and Petrochemicals, Coal and Mining, General and Heavy Engineering.



Figure: Illustrative Steel Slabs

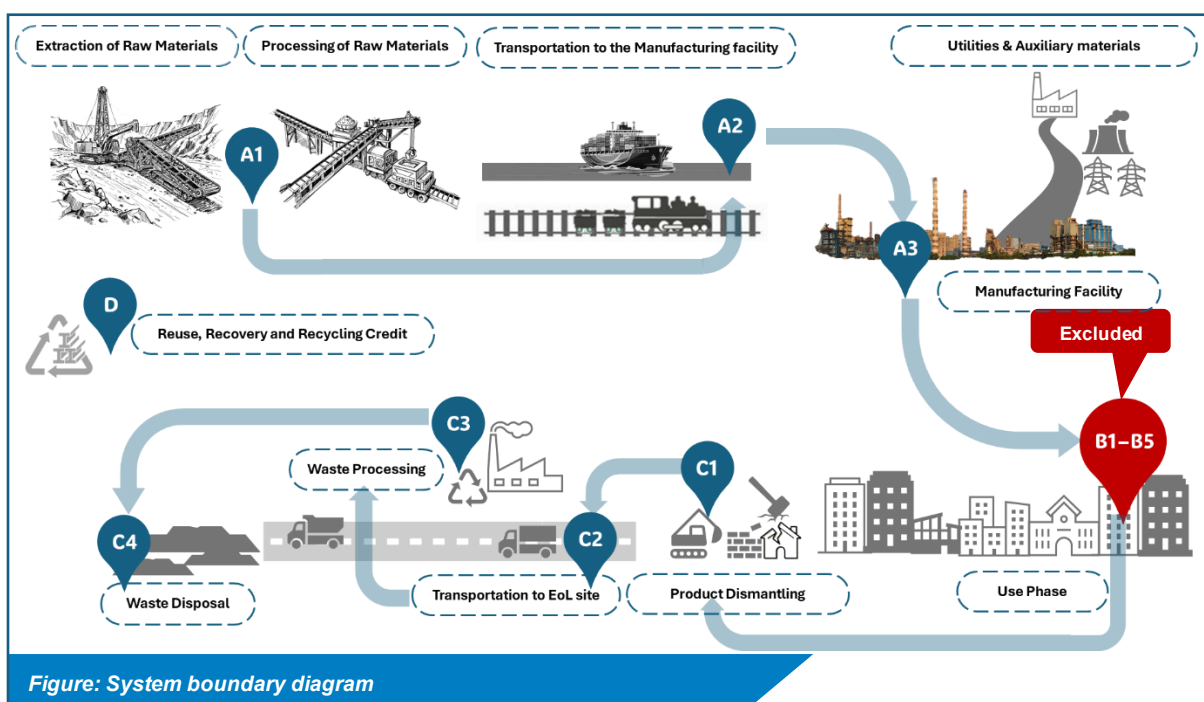
Product Range (Size and Grade details)

Manufacturing Unit	Product	Size	Grades
Jamshedpur	Steel Slab	200- 220 mm (Thickness) 900- 1550 mm (Width)	Ultra-low IF Steel, High Carbon Steel, Low Carbon Steel, Medium Carbon Steel, Electrical Steel,
Kalinganagar		210- 250 mm (Thickness) 900- 2100 mm (Width)	
Meramandali		210- 250 mm (Thickness) 900- 1680 mm (Width)	

Life Cycle Assessment

LCA information

Functional unit / declared unit	1 Tonne (1000 kg) Tata Steel Average Steel slab
Reference service life	Not applicable
Time representativeness	The collection of foreground data refers to April 2022 to March 2023
Database(s) and LCA software used	The LCA model was created using Sphera's LCA for Experts (LCA FE) software, version 10.9
Description of system boundaries	Cradle to gate with modules C1–C4 and module D (A1–A3 + C + D)
Reference package used	EN 15804 reference package based on EF 3.1



Module A1 – A3

The product stage includes provision of all materials, products and energy, as well as waste processing up to the end-of waste state or disposal of final residues during the product stage. The electricity in Tata Steel sites have been sourced from a mix of Grid (76%), Captive

power plant (23%), and renewable sources (<1%) mostly solar and the emission factor of the electricity mix for the GWP-GHG indicator is 1.35 kg CO₂eq./kWh. There is no packaging considered in the present assessment.

Module C1 – C4

Within this EPD, the modules C1-C4 are included. These modules consider the dismantling of the considered product (C1), the transportation of the dismantled components to their End of Life (EoL) destination (C2), the waste processing for recovery or recycling (C3)

as well as the disposal (C4), if given. At EoL, the steel material leaves the product system in C3 for recycling in Module D. The considered EoL scenario for the steel material is 88% recycling and 12% landfill.

Category	Subcategory	Unit	Quantity
Collection process	Collected separately	kg	1000
	Collected with mixed construction waste	kg	0
	Reuse	kg	0
Recovery	Recycling	Kg	880
	Landfill	Kg	0
	Incineration	Kg	0
	Incineration with energy recovery	Kg	0
	Energy conversion efficiency rate	Kg	0
Disposal	Material for final disposal	kg	120
Transport	Deconstruction site to scrap processing plant	Km	0
	Scrap processing plant to site for end of waste	km	50

Module D

Module D includes any declared benefits and loads from net flows leaving the product system that have not been allocated as co-products and that have passed the end-of-waste state in the form of reuse, recovery and/or recycling potentials. Metals are assumed to reach the end of waste status directly at the construction site. The treatment as well as net benefits and

loads of reuse or recycling potentials (for the net scrap amount only) are grouped to module D. In the EoL net scrap approach (Amount of steel recycled at EoL – Scrap input from previous product life cycles) has been used. The environmental burden for the extra scrap used in the manufacturing has been considered in module D.

Data quality assessment and declaration

Process	Source type	Source	Reference year	Data category	Share of primary data, of GWP-GHG results for A1-A3
Manufacturing of product	Collected data	EPD owner	FY 2023	Primary data	75.43%
Generation of electricity used in manufacturing of product	Collected data and database	Sphera MLC 2024.2 and EPD owner	2024	Primary data	12.17%
Transport of raw materials	Database	Sphera MLC 2024.2	2024	Primary data	2.04%
Total share of primary data, of GWP-GHG results for A1-A3					89.64%

Note: The share of primary data is calculated based on GWP-GHG results. It is a simplified indicator for data quality that do not capture all relevant aspects of data quality. The indicator is not comparable across product categories.

Cut-off Criteria

The environmental impact of the product studied has been assessed by considering all significant processes, materials, and emissions. Excluded flows are assumed to have a negligible impact, contributing less than

5% to the cumulative impact assessment categories. The production of capital equipment, facilities, and infrastructure required for manufacture has not been considered.

Data quality and sources

Data quality is compliant with ISO 14025:2006. All primary data were collected for the financial

year 2023. All background data come from the Sphera MLC 2023.2.

Allocation

With any multi-product system, allocation rules are defined to relate the system inputs and outputs to each of the products. Several methods are documented in ISO 14040:2006 and ISO Technical Report 14049.

Modules declared, geographical scope, share of specific data (in GWP-GHG results) and data variation (in GWP-GHG results)

Modules of the production life cycle included (X- Module declared, ND- Not declared)

	Product Phase			Construction process stage		Use stage							End of Life stage				Resource recovery stage
	Raw material supply	Transport	Manufacturing	Transport	Construction Installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
Module	A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
Declared	X	X	X	ND	ND	ND	ND	ND	ND	ND	ND	ND	X	X	X	X	X
Geography	India												Global				Global
Specific data used	89.6%																
Variation – products	<1%																
Variation – sites	< 10%																

Material properties in Average Steel slab

Product components	Weight, kg	Post-consumer material, Weight (%)	Biogenic material, weight % of product	Biogenic material, kg C/product or declared unit
Steel	1000 kg	4	0	0
Iron	>98.5%	-	0	0
Carbon	0.001-0.5%	-	0	0
S	0-0.02%	-	0	0
P	0-0.02%	-	0	0
Mn	0-2%	-	0	0
Total	100%	-	0	0

The packaging material is having minimal impacts, so it has not been considered in the assessment. So, the biogenic content of the packaging material has not been mentioned in the table. Products do not contain any substances that can be included in “Candidate List of Substances of Very High Concern for Authorization” and raw materials used are not part of the EU REACH regulation.

Results of the environmental performance indicators

Mandatory impact category indicators according to EN 15804+A2:2019

Results per 1 Tonne of Average Steel Slab							
Indicator	Unit	A1 - A3	C1	C2	C3	C4	D
GWP-total	kg CO ₂ eq.	2.45E+03	0.00E+00	5.36E+00	0.00E+00	3.57E+00	-1.45E+03
GWP-fossil	kg CO ₂ eq.	2.45E+03	0.00E+00	5.36E+00	0.00E+00	3.57E+00	-1.45E+03
GWP-biogenic	kg CO ₂ eq.	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
GWP-lulc	kg CO ₂ eq.	3.89E-01	0.00E+00	1.21E-04	0.00E+00	1.45E-02	-1.94E-01
ODP	kg CFC 11 eq.	2.75E-09	0.00E+00	1.02E-13	0.00E+00	1.04E-11	-1.96E-09
AP	mol H ⁺ eq.	6.66E+00	0.00E+00	4.14E-02	0.00E+00	2.08E-02	-3.57E+00
EP-freshwater	kg P eq.	2.23E-04	0.00E+00	8.18E-07	0.00E+00	6.74E-06	-3.4E-04
EP-marine	kg N eq.	1.14E+00	0.00E+00	2.06E-02	0.00E+00	5.03E-03	-5.73E-01
EP-terrestrial	mol N eq.	1.25E+01	0.00E+00	2.26E-01	0.00E+00	5.53E-02	-4.89E+00
POCP	kg NMVOC eq.	3.38E+00	0.00E+00	3.98E-02	0.00E+00	1.58E-02	-2.22E+00
ADP-minerals & metals*	kg Sb eq.	8.21E-04	0.00E+00	6.30E-08	0.00E+00	2.23E-07	-7.87E-03
ADP-fossil*	MJ	2.12E+04	0.00E+00	7.00E+01	0.00E+00	5.53E+01	-1.38E+04
WDP*	m ³	1.19E+02	0.00E+00	6.56E-03	0.00E+00	4.08E-01	-9.38E+01
Acronyms	GWP - total = global warming potential; GWP - fossil = global warming potential (fossil fuel only); GWP - biogenic = global warming potential (biogenic); GWP - lulc = global warming potential (land use only); ODP = ozone depletion; AP = acidification terrestrial and freshwater; EP freshwater = eutrophication potential (freshwater); EP - marine = eutrophication potential (marine); EP- terrestic = eutrophication potential (terrestrial); POCP = photochemical ozone formation; ADPE = abiotic depletion potential (element); ADPF = abiotic depletion potential (fossil); WDP = water scarcity						

Additional mandatory and voluntary impact category indicators

Results per 1 Tonne of Average Steel Slab							
Indicator	Unit	A1-A3	C1	C2	C3	C4	D
GWP-GHG*	kg CO ₂ eq.	2.45E+03	0.00E+00	5.36E+00	0.00E+00	3.57E+00	-1.45E+03
Acronyms	GWP - GHG*= global warming potential (greenhouse gases) This indicator includes all greenhouse gases included in GWP-total but excludes biogenic carbon dioxide emissions and uptake and biogenic carbon stored in the product with characterization factors (CFs) based on IPCC (2013)						

Resource use indicators according to EN 15804+A2:2019

Results per 1 Tonne of Average Steel slab							
Indicator	Unit	A1-A3	C1	C2	C3	C4	D
PERE	MJ	7.25E+02	0.00E+00	1.44E-01	0.00E+00	8.20E+00	-5.73E+02
PERM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PERT	MJ	7.25E+02	0.00E+00	1.44E-01	0.00E+00	8.20E+00	-5.73E+02
PENRM	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
PENRE	MJ	2.12E+04	0.00E+00	7.00E+01	0.00E+00	5.53E+01	-1.45E+04
PENRT	MJ	2.12E+04	0.00E+00	7.00E+01	0.00E+00	5.53E+01	-1.45E+04
SM	kg	4.00E+01	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
RSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
NRSF	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
FW	M ³	3.05E+00	0.00E+00	2.02E-04	0.00E+00	1.23E-02	-2.0E+00
Acronyms	PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy re-sources; SM: Use of secondary materials; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water						

Output flows & Waste indicators according to EN 15804+A2:2019

Results per 1 Tonne of Average Steel slab							
Indicator	Unit	A1-A3	C1	C2	C3	C4	D
HWD	kg	1.37E-05	0.00E+00	2.38E-05	0.00E+00	1.32E-08	-1.01E-05
NHWD	kg	1.54E+00	0.00E+00	6.35E+00	0.00E+00	1.60E+02	-1.52E+00
RWD	kg	3.95E-02	0.00E+00	1.49E-01	0.00E+00	7.50E-04	-1.51E-03
MFR	kg	0.00E+00	0.00E+00	0.00E+00	8.80E+02	0.00E+00	0.00E+00
MER	Kg	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EEE	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
EET	MJ	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00	0.00E+00
Acronyms	HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed; MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy						

Additional environmental performance indicators according to EN15804+A2:2019

Results per 1 Tonne of Average Steel slab							
Indicator	Unit	A1-A3	C1	C2	C3	C4	D
PM	Disease incidences	9.22E-05	0.00E+00	2.62E-04	0.00E+00	2.40E-07	-3.35E-05
IR	kBq U235 eq.	3.68E+00	0.00E+00	1.39E+01	0.00E+00	1.03E-01	-2.01E+00
ETF	CTUe	3.82E+03	0.00E+00	1.08E+04	0.00E+00	3.36E+01	-7.55E+01
HTP-c	CTUh	5.18E-08	0.00E+00	2.17E-07	0.00E+00	7.32E-10	-4.46E-08
HTP-nc	CTUh	2.19E-06	0.00E+00	8.48E-06	0.00E+00	2.60E-08	-1.53E-06
SQP	Pt	6.90E+02	0.00E+00	2.83E+03	0.00E+00	1.01E+01	-3.43E+02
Acronyms	PM = Particulate matter emissions; IR = Ionising radiation, human health; ETF= Eco-toxicity (freshwater); HTP-c = Human toxicity, cancer effects; HTP-nc = Human toxicity, non-cancer effects; SQP = Soil quality potential/Land use related impacts						

* Disclaimer: The results of this environmental impact indicator shall be used with care as the uncertainties of these results are high or as there is limited experience with the indicator.

Reference

- General Programme Instructions of the International EPD® System. Version 4.0.
- PCR 2019:14. Construction Products, Version 1.3.4 (Valid upto: 2025-06-20)
- Sustainability of construction works - Environmental product declarations - Methodology for selection and use of generic data; CEN/TR 15941:2010
- EN 15804: EN 15804:2012+A2:2019: Sustainability of construction works -Environmental Product Declarations - Core rules for the product category of construction products.
- EN ISO 14025: EN ISO 14025:2011-10 Environmental labels and declarations - Type III environmental declarations - Principles and procedures
- EN ISO 14040: EN ISO 14040:2009-11 Environmental management - Life cycle assessment - Principles and framework
- LCA FE: LCA FE Software System and Database for Life Cycle Engineering, Sphera Solution GmbH, Leinfelden-Echterdingen, 2024 (<https://sphera.com/solutions/product-stewardship/life-cycle-assessment-software-and-data/managed-lca-content/>)
- WSI and Eurofer's Co-Product Allocation Methodology 2014 – A methodology to determine the LCI of Steel Industry Co-products
- World Steel Association- CO2 Data Collection User Guide, Version 9 (May 2019).
- EN ISO 14044: EN ISO 14044:2006-10 Environmental management - Life cycle assessment - Requirements and guidelines

